25 A copper disc of area A rotates at angular frequency ω at the centre of a long solenoid of length L and having N turns. The solenoid carries a current I. The plane of the disc is normal to the magnetic flux. The rotation rate is adjusted so that the e.m.f. generated between the centre of the copper disc and its rim is 10% of the potential difference across the ends of the solenoid.

Which expression gives the potential difference across the ends of the solenoid?

A $10\mu_o NIA\omega$
B $1.6\mu_o NIA\omega / L$
C $0.1\mu_o NIA\omega$
D $0.016\mu_o NIA\omega / L$